# Muyao Kong

+1 (914) 893-7586 | mk9014@nyu.edu | https://muyao-kong.github.io/

## EDUCATION

**New York University** New York, NY
*Major in Computer Science* Expected May 2027
*Minor in Mathematics*
**Honors**: Dean's List
**Relevant Coursework**: Natural Language Processing, Basic Algorithm, Linear Algebra, Probability & Statistics, Discrete Math, Data Structures

**High School Attached to Shandong Normal University** Jinan, China
*GPA: 4.01 Ranking: 5%* Graduated June 2023

## RESEARCH

**Read the Room Before Your Model Generates**
Project Lead and Researcher

- Proposed **Scenario-Actor**, a prompt-level framework that selects LLM decoding temperature from the communicative scenario before generation.
- Designed **Scenarize** to map r-th layer Transformer hidden states into an interpretable **scenario embedding** representing conversational context.
- Designed **Actout**, a scenario-axis readout function that maps scenario embeddings to scalar decoding temperatures.
- Designed a triplet-supervised training setup to learn scenario embeddings from relative differences among communicative contexts.

**Generate-then-Rerank for Pedagogically Calibrated Language-Model Explanations**
Researcher

- Applied a **generate-then-rerank** pipeline to improve perceived-grade alignment of LLM explanations across Elementary, High School, and Graduate levels.
- Built a target-aware reranker combining linguistic register features with a fine-tuned **DistilBERT** perceived-grade classifier.
- Co-designed interpretable register features capturing lexical, rhetorical, and discourse-level differences in audience-calibrated explanations.
- Evaluated reranking with paired-rank accuracy and an end-to-end annotation study, finding directional improvements for High School and Graduate explanations.

## PUBLICATION & MANUSCRIPT

[1] **Muyao Kong**, and Zhihao Lin. "Read the Room Before Your Model Generates." *Under review at the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.
[2] Muqiao Tao, **Muyao Kong**, and Qitian Xiong. "Generate-then-Rerank for Pedagogically Calibrated Language-Model Explanations." 2026 (manuscript in preparation).

## WORK EXPERIENCE

**National Supercomputing Center** Jinan, China
Algorithm and Data Analyst Intern Jun 2024 - Aug 2024

- **Cleaned** time series device data; constructed data anomaly detection and localization algorithms to locate and process abnormal data data using **Huber Regressor** in **robust regression**.
- Built **ICC (Intraclass Correlation)** data consistency check algorithms, made judgment of consistency on the historical data of multiple related devices.
- Built machine learning **ARIMA** (Autoregressive Integrated Moving Average) model, used the constructed ARIMA model to forecast the **time series** device data.
- Wrote Python automation script for timed processing of device data, handled timed database read/write using **PyMySQL**, and processed the retrieved device data from MySQL database.

**China Cinda Asset Management Co., Ltd** Beijing
Risk Management Intern Jun 2025 – Aug 2025

- Synthesized industry and internal case materials on **distressed asset resolution** (e.g., **debt-to-equity swaps, auctions, restructurings**) and distilled key risk factors into reusable briefings for internal reference.
- Compiled and standardized annual report data across 18 financial institutions including banks, AMCs and brokers, extracting and consolidating key financial and disclosure items into structured datasets for analysis.
- Performed basic **cross-company** and **time-series comparisons** (leverage, liquidity, profitability quality, impairments, contingencies) and drafted concise risk highlights to support internal documentation.
- Operated under strict data-access and compliance constraints, focusing on public disclosures and documentation workflows to improve traceability and retrieval efficiency.

## TECHNICAL SKILLS

**Programming Languages:** Proficient - Python, Java, C++, JavaScript, Familiar - C, Assembly Language
**Markup Languages:** Proficient - LaTeX, HTML, CSS
**Other Libraries:** Proficient - Pandas, NumPy, Matplotlib, Familiar - Statsmodel, PyTorch

## OTHER SKILLS

**Languages:** Mandarin, English, Japanese, Arabic
**Interests:** Artificial Intelligence, Mathematics, Foreign Languages, Linguistics, Economics, Piano